

P3868R1

Allow `#line` before module declarations

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This version:

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Abstract

[P1857R3] disallowed any control-lines before the global module fragment. This has turned out to be too strict due to many common tools running the preprocessor separately. We need to allow `#line` directives on any line. This resolves US 55-102.

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§ 1. Changelog

- R1 - Change wording to not use a *group*, and be compatible with Alisdair's [editorial change to the line directive wording](#).

§ 2. The Problem

After implementing part of [P1857R3], [Clang soon discovered](#) that it makes the following code ill-formed:

```
#line 1 "A.cppm"  
export module a;
```

This is a problem because tools such as [build2](#), [ccache](#), and [distcc](#), among many others, rely on feeding the preprocessed output back into the compiler. This output generally contains `#line` (or GNU line marker) directives at the start of the file.

§ 3. The Fix

The standard should allow `#line` directives to appear on any line.

§ 4. Wording

Modify [cpp.pre]:

```
module-file:  
    line-directivesopt pp-global-module-fragmentopt pp-module groupopt pp-private-module-fragmentopt
```

control-line:

```
# include pp-tokens new-line
pp-import
line-directive
# embed pp-tokens new-line
# define identifier replacement-list new-line
# define identifier lparen identifier-listopt ) replacement-list new-line
# define identifier lparen . . . ) replacement-list new-line
# define identifier lparen identifier-list , . . . ) replacement-list new-line
# undef identifier new-line
# line pp-tokens new-line
# error pp-tokensopt new-line
# warning pp-tokensopt new-line
# pragma pp-tokensopt new-line
# new-line
```

line-directives:

line-directive line-directives_{opt}

Add before the first paragraph in [cpp.line]:

line-directive:

~~# *line pp-tokens new-line*~~

§ References

§ Informative References

[P1857R3]

Michael Spencer. *Modules Dependency Discovery*. 11 February 2020. URL:
<https://wg21.link/p1857r3>